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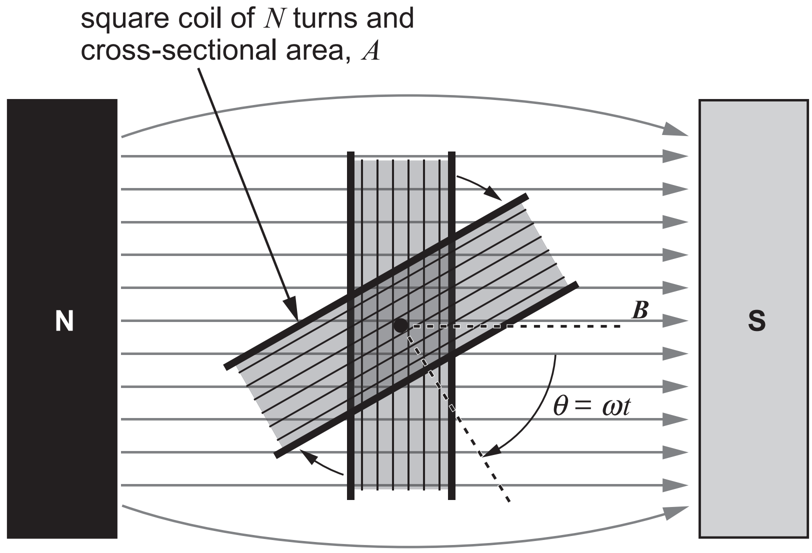
Option A – **Alternating Currents**

**6.** *(a)* A coil is rotated in a magnetic field.




(i) Use Faraday's law to explain why the emf is proportional to the angular velocity of the coil. [2]

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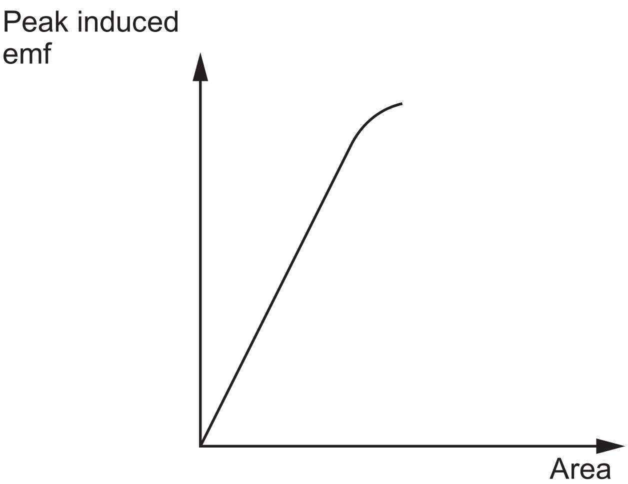
The peak emf induced in a rotating coil is given by:

$$E = \omega BAN$$

An experiment is carried out to investigate the variation in the peak emf with area, $A$, by using coils with different cross-sectional areas. The following graph is obtained.





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(ii) Discuss the agreement of the graph with the equation for the peak emf and suggest a reason for any disagreement (see diagram of coil opposite). [3]

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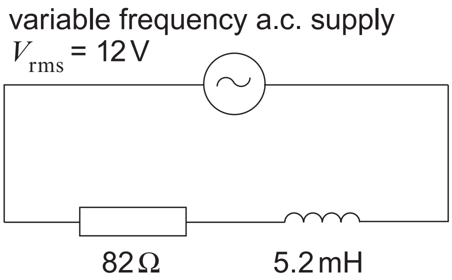
*(b)* (i) The minimum impedance of the circuit below occurs when the frequency of the a.c. supply is zero. Explain why this is so. [2]

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(ii) Calculate the rms current when the rms pd across the resistor and inductor are equal. [3]

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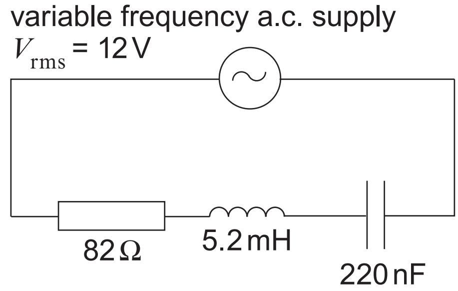
*(c)* (i) Explain how resonance arises in the $LCR$ circuit shown. [3]




(ii) Calculate the power dissipated in the circuit at resonance. [2]

(iii) Calculate the rms current when the frequency of the supply is 9.40 kHz. [3]

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(iv) Discuss briefly the effectiveness of the given $LCR$ circuit for providing a sharp resonance curve. [2]

(It may be useful to note that $\frac{\omega_0 L}{R} = \frac{1}{R}\sqrt{\frac{L}{C}}$.)

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